

Markov Processes: Homework 5

Due Wed Oct 7 by 11:59pm on Gradescope

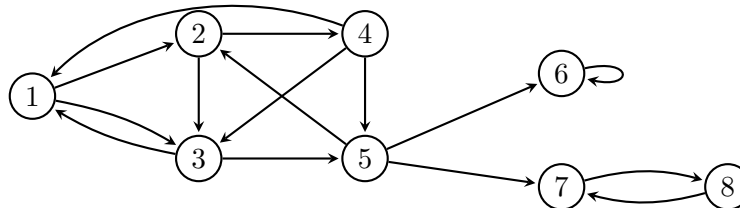
Format. Plots, code, and write-up follow the conventions of Homeworks 1–4.

Optional problems. Everything after the END box is practice and is not graded.

1. **Stationary, but not converging: a four subunit channel.** The Hodgkin–Huxley potassium channel (*J. Physiol.* **117**, 500, 1952) conducts only when all four of its gating subunits are active, the origin of the n^4 in their conductance. Let $X_n \in \{0, \dots, 4\}$ count the active subunits. Each step one subunit is chosen uniformly; if inactive it activates with probability a , if active it deactivates with probability b . Thus, with $0 < a, b \leq 1$,

$$p(k, k+1) = a \frac{4-k}{4}, \quad p(k, k-1) = b \frac{k}{4}, \quad p(k, k) = 1 - p(k, k+1) - p(k, k-1).$$

- (a) Use detailed balance between neighbors, $\pi(k)p(k, k+1) = \pi(k+1)p(k+1, k)$, to show that π is binomial($4, \theta$) with $\theta = a/(a+b)$, and explain why it is the only stationary distribution.
 - (b) Take $a = b = 1$ and $q_0 = \delta_0$. Compute $q_n = q_0 p^n$ for $n \leq 60$, report q_{50} and q_{51} , and plot $q_n(2)$, $q_n(4)$ and the running average $\frac{1}{n+1} \sum_{k \leq n} q_k(2)$. Explain why $p^n(k, k) = 0$ for odd n , and how π can be unique while q_n fails to converge.
 - (c) Now take $a = 0.3$, $b = 0.1$. Compute π by a linear solve and check it against part (a). Add q_n from δ_0 as a second panel, and report the smallest n with $\max_k |q_n(k) - \pi(k)| < 10^{-6}$ and the second largest eigenvalue modulus of p . What did the self loops do?
2. **Who is the champion? Detailed balance and its failure.** Players $1, \dots, m$ play head to head, and i beats j with probability $A_{ij} = 1 - A_{ji}$. The *champion chain* records the current champion. Each step a challenger is chosen uniformly from the other $m-1$ players and the winner becomes champion, so $p(i, j) = A_{ji}/(m-1)$ for $j \neq i$, and $p(i, i) = 1 - \sum_{j \neq i} p(i, j)$.
 - (a) In the Bradley–Terry model $A_{ij} = s_i/(s_i + s_j)$ with strengths $s_i > 0$. Show that $\pi_i = s_i / \sum_k s_k$ satisfies detailed balance for every m . For $m = 3$ and $s = (3, 2, 1)$, write p and give the mean return time to the weakest player.
 - (b) Show that detailed balance with all $\pi_i > 0$ implies $p(1, 2)p(2, 3)p(3, 1) = p(1, 3)p(3, 2)p(2, 1)$ (Kolmogorov’s criterion). Now take a rock–paper–scissors league, where A beats B , B beats C and C beats A , each with probability $2/3$. Write p , show it is doubly stochastic, give π , show detailed balance fails, and compute the net flow $\pi_i p(i, j) - \pi_j p(j, i)$ around the cycle.
 3. **Simulation Assignment: PageRank and the random surfer.** Brin and Page (*Comput. Netw. ISDN Syst.* **30**, 107, 1998) ranked pages by the stationary distribution of a surfer who follows a uniformly chosen outgoing link. Groups of pages that link only to each other trap the surfer, so they let the surfer follow a link with probability d and jump to a uniform random page with probability $1 - d$. Consider the web below.



- (a) Write the 8×8 matrix p with no random jumps, list its closed classes, and find all stationary distributions. From $q_0 = \frac{1}{8}(1, \dots, 1)$ compute q_{100} and q_{101} , and name the two ways p fails to give a ranking.
- (b) Form $G = dp + \frac{1-d}{8} \mathbf{1}\mathbf{1}^T$ with $d = 0.85$. If q_n and π are probability vectors with $\pi G = \pi$, show that $q_{n+1} - \pi = d(q_n - \pi)p$, and conclude that $\|q_n - \pi\|_1 \leq d^n \|q_0 - \pi\|_1$, using $\|vp\|_1 \leq \|v\|_1$ for stochastic p . Why does this make π unique? Compute π by power iteration from uniform

until $\|q_{n+1} - q_n\|_1 < 10^{-10}$, check it with a linear solve, and report the iteration count, π and the ranking. Why do pages 1 and 5 tie?

- (c) Run $R = 100$ surfers on G for $T = 10^5$ steps from page 1, sampling each step by the discrete inverse transform of Homework 2, vectorized over surfers. Estimate π by time averages, show one surfer's estimate beside π in a bar chart, and plot the rms over surfers of $\max_i |\hat{\pi}_i - \pi_i|$ against T on log-log axes with a fitted slope over $10^2 \leq T \leq 10^5$. Why is the error at $T = 10^5$ several times larger than $\sqrt{\pi_6(1 - \pi_6)/T}$?

END

4. **Optional Practice: random walks on graphs.** A random walk on a finite, connected, undirected graph moves from x to a uniformly chosen neighbor, so $p(x, y) = 1/\deg(x)$ for neighbors.
- In lecture, $\pi(x) = \deg(x)/(2|E|)$ was checked directly and through the edge chain. Show that it also satisfies detailed balance. For which graphs is p itself doubly stochastic?
 - A king moves on an 8×8 chessboard to a uniformly chosen adjacent square (up to eight). Find $2|E|$ and the mean return times to a corner, an edge square, and an interior square.
 - For the directed graph $1 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 3, 3 \rightarrow 1$, with uniformly chosen outgoing edges, find π and show it is proportional to neither the in- nor the out-degrees. Why does part (a) fail?
5. **Optional Practice: two facts about detailed balance.**
- Show that the stationary distribution of every finite birth-death chain on $\{0, \dots, N\}$ satisfies detailed balance. *Hint.* Sum the equations $(\pi p)(j) = \pi(j)$ over $j \leq k$ and compare the probability flow across the cut between k and $k + 1$.
 - Suppose p satisfies detailed balance with respect to π , with all $\pi(x) > 0$, and let D be the diagonal matrix with entries $\pi(x)$. Show that $S = D^{1/2}pD^{-1/2}$ is symmetric, and conclude that every eigenvalue of p is real. Check this against the spectra in Problems 1 and 2.
6. **Optional Practice: an epidemic with spontaneous infection.** In the discrete time epidemic of Lecture 18, $X_n \in \{0, \dots, N\}$ counts the infected. Each step a new infection occurs with probability $p_n = \beta n(N - n)\Delta t$ for $1 \leq n < N$ and a recovery with probability $q_n = \gamma n\Delta t$. With nobody infected, an outside infection arrives with probability $p_0 = \varepsilon$. Take $N = 10$, $\beta = 1/N^2$, $\gamma = 1/N$, $\varepsilon = 0.1$ and $\Delta t = 0.1$.
- Find π by detailed balance, report it to four digits along with the mean number infected, and find the mean time between successive outbreaks, meaning returns to $X = 0$. Then repeat with $\beta = 3/N^2$ and describe how the shape of π changes.
 - Now set $\varepsilon = 0$. Find every stationary distribution, and explain why the answer of part (a) says nothing useful about what an observer would see over any reasonable time.
7. **Optional Practice: choosing the damping factor.**
- Repeat Problem 3(b) with $d = 0.5$ and $d = 0.99$. Report π and the iteration counts against $\ln(10^{-10})/\ln d$. What happens to the rank of pages 1–5 as $d \rightarrow 1$?